SOL: Structural Operational Language

Engineering Grounding Protocol for AI-Assisted Mechanical Analysis

Version 1.2 — Experimental Specification

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2026

Abstract

Structural Operational Language (SOL) is a semantic protocol designed to stabilize technical context among heterogeneous agents, including human engineers, large language models (LLMs), CAD systems, and programmable logic controllers (PLCs). By transforming natural language descriptions into structured causality graphs, SOL establishes an intermediate engineering representation that preserves topology, causality, and constraints. SOL serves as a reasoning stabilization layer that preserves engineering intent across human-AI boundaries. It is not intended to replace CAD kernels, FEA tools, or physics simulators.

Contents

1 Problem Statement

Current engineering workflows encounter critical reliability barriers when integrating AI agents into technical reasoning chains:

- **Semantic drift:** The gradual loss or mutation of engineering meaning, topology, constraints, or functional intent during extended natural-language reasoning chains.
- **CAD intent gap:** Geometric modeling systems store exact spatial data but systematically fail to capture causal intent, operational failure modes, or thermodynamic boundaries.
- **Long-horizon inconsistency:** Autonomous agents struggle to maintain causal coherence regarding physical systems across multi-step diagnostic sequences.
- **Multi-agent coordination:** Heterogeneous systems lack a shared grounding representation, forcing semantic loss at integration boundaries.

2 Core Concepts

SOL provides a minimal semantic vocabulary visually scannable by humans and deterministically parsable by machines. The protocol operates strictly as a cognitive grounding layer and causal topology map.

2.1 Basic Syntax Philosophy

To maximize cross-cultural accessibility and parsing determinism, SOL notation intentionally mimics school-level mathematics and common engineering shorthand. The core grammar relies on four foundational structures:

- `[]` = Entities and physical objects (e.g., `[BRG*:6204]`).
- `()` = Actions, processes, or system states (e.g., `(ANALYZE)`).
- `:` = Specification, parameterization, or classification.
- `->` = Flow, causality, sequence, or load vector.

3 Token Specification

3.1 Entity Tokens

Entity tokens `[...]` represent physical objects or media existing within the system boundaries.

Token	Meaning	Example
`[MAT]`	Material	`[MAT:FE/45]`, `[MAT:AL/6061]`
`[BRG*]`	Rolling bearing (ball, roller, needle)	`[BRG*:6204]`, `[BRG*:NU206]`
`[BUS]`	Plain bearing / bushing	`[BUS:bronze]`, `[BUS:PTFE]`
`[SPG]`	Spring / elastic element	`[SPG:COMPRESSION/k=5.2]`
`[SFT]`	Shaft / rotational axis	`[SFT:d=20]`
`[FLU]`	Fluid or gas medium	`[FLU:OIL]`, `[FLU:AIR]`
`[MTR]`	Motor / energy converter	`[MTR:AC]`, `[MTR:DC]`
`[SENS]`	Sensor / telemetry node	`[SENS:TEMP]`, `[SENS:VIB]`

Table 1: Core Entity Primitives. The marker * denotes rolling contact.

3.2 Operator Tokens

Operator tokens `(...)` define relationships, diagnostic processes, and dynamic system states.

Operator	Meaning	Use
`->`	Causality / Transfer	Energy, material, signal propagation
`(ANALYZE)`	Analysis / diagnostics	FFT, inspection, signal processing
`(WARN)`	Anomaly / warning	Threshold violation, instability
`(STOP)`	Block / emergency stop	Constraint failure, safety limit
`(REPAIR)`	Action / maintenance	Calibration, replacement
`(OK)`	Stabilization / anchor	Validated equilibrium state
`(ERR:...)`	Physics violation	Conservation law failure

Table 2: Operator Primitives

3.3 Operator Notation: Canonical and Human Layers

The protocol deliberately fragments operator notation into two distinct registers to separate machine determinism from human cognition:

- **Canonical operators** (Mandatory for SOL-A, SOL-IR): Text-based, deterministic ASCII strings. Examples: `(ANALYZE)`, `(WARN)`, `(OK)`.

- **Human-layer shorthand** (Optional, SOL-H only): Rendering-dependent emoji glyphs functioning strictly as cognitive overlays. These accelerate human visual parsing during diagnostic reviews but remain formally non-canonical.

SOL-H (Optional Glyph)	Canonical Token (Authoritative)
□ (magnifier)	`(ANALYZE)`
□ (warning sign)	`(WARN)`
□ (stop sign)	`(STOP)`
□ (wrench)	`(REPAIR)`
□ (anchor)	`(OK)`
□ (handshake)	`(SYNC)`

Table 3: Emoji shorthand mapping. Glyph rendering depends entirely on the host environment.

3.4 Specification Syntax

Tokens accept nested parameterization to inject engineering specificity directly into the topology:

```
[BRG*:6204]              % standard type
[BRG*:6204/ISO15]        % standard type + normative reference
[BRG*:6204/d=20/D=47]    % explicitly parameterized
[MTR:AC/P=5kW/RPM=1450]  % parametric energy converter
[SENS:TEMP/PT100]        % explicitly typed telemetry
```

4 Architectural Layers

SOL is constituted by four architectural layers, each carrying distinct invariances.

Layer	Identity	Core Function
L1: SOL-H	Cognitive Acceleration	Human-facing visual notation. Permits emoji shorthand to reduce friction in diagnostic chat and maintenance logs.
L2: SOL-A	Inter-Agent Interchange	Strict, stable protocol establishing physical invariants between heterogeneous systems (LLMs, PLCs, expert systems).
L3: SOL-IR	Internal Reasoning Substrate	Mutable cognitive constitution for AI agents. Systems may compress or extend this graph internally to compute causal chains.
L4: SOL-CAD/DW	Topological Grounding Layer	Formal mapping of spatial relationships, geometric constraints, and mechanical fits relative to coordinate axes.

Table 4: SOL Architectural Layers

SOL-CAD/DW represents spatial grounding (`[FIT:H7/k6]`, `[AX:CENTER]`). It does not compute geometric intersections or replace CAD kernels; it extracts topological dependencies required for logical reasoning.

5 Usage Examples

5.1 Bearing Vibration Diagnostics

Bearing vibration monitoring chain

```
[BRG*:6204] -> [SENS:VIB] -> □ (ANALYZE):FFT
          -> □ (WARN):IMBALANCE -> □ (REPAIR):ALIGN -> □ (OK)
```

Interpretation: A specific rolling bearing outputs vibration telemetry; FFT analysis identifies an imbalance state; a physical alignment action corrects the anomaly; the system anchors into a validated equilibrium.

5.2 Hydraulic Leak Detection

Hydraulic system leak detection

```
[FLU:OIL] -> [PMP] -> [SENS:PRES]
          -> □ (ANALYZE) -> □ (WARN):LEAK -> [VAL] -> □ (CLOSE) -> □ (OK)
```

5.3 Full Industrial Chain with Standards

Industrial drive with parametric specification

```
[MTR:AC/P=15kW/IEC60034]
  -> [CPL:FLEX/DIN740]
  -> [SFT:FE40X/d=45]
  -> [BRG*:7210B/ISO15]
  -> [GR:WORM/i=50]
  -> [FLU:OIL/ISO_VG_220]
  -> [SENS:TEMP/PT100]
  -> □ (ANALYZE):T_GRADIENT
  -> □ (WARN):THERMAL_RUNAWAY -> □ (STOP)
```

6 Engineering Rationale

6.1 Responsibility Axiom

Every element mapped within a SOL protocol string carries deterministic physical responsibility. A vector `->` signifies a discrete load transfer, a fluid clearance, or a signal path. It never functions as a decorative or ambiguous referent.

6.2 Observability Requirement

Any mechanical reasoning chain lacking `[SENS]` (telemetry) nodes is technically classified as incomplete for automated diagnostics. Causal paths governing reliability require explicit observability anchors.

6.3 Invariant Verification

SOL reasoning chains must structurally conserve thermodynamic and physical laws. A causal topology that spontaneously generates energy without a defined source resolves to a hard (ERR:PHYSICS_VIOLATION

6.4 Ontological Discipline

Tokens demand precise physical compliance. The * marker in [BRG*] physically restricts the entity to rolling-element kinematics. Sliding friction interfaces mandate the [BUS] token. Conflating these typologies constitutes an ontological reasoning error.

7 Limitations

The current experimental specification operates within defined boundaries:

- **Continuous deformation:** Elasticity, thermal expansion, and plasticity are not captured by discrete token chains, necessitating future continuous-state extensions.
- **Nonlinear transitions:** The protocol maps discrete causality. Chaotic systems and phase bifurcations fall outside the current envelope.
- **CAD integration:** While SOL-CAD/DW defines the mapping syntax, autonomous extraction of these semantic graphs directly from native CAD geometry kernels remains an unsolved research challenge.
- **Time domain:** SOL dictates causal topology, not temporal kinetics. Transient delays and thermal lags require external simulators attached via `(LAW:...)` protocol extensions.

8 Future Work

8.1 Proposed Validation Methodology

Formal empirical validation of the semantic drift hypothesis remains pending. Proposed benchmarking will execute identical complex engineering diagnostics through heterogeneous multi-agent systems, measuring outputs across:

- Unrestricted natural language prompts.
- SOL-anchored causality graphs.

The primary metrics include topological preservation accuracy, causal consistency throughout failure tree resolution, and the rate of missing-component hallucination.

8.2 Infrastructure Development

- Publication of an open, machine-readable invariants library mapping standard diagnostic chains (e.g., `[TOKEN] -> (ANALYZE) -> [SENS]`) to ISO specifications.
- Deployment of an L3 (SOL-IR) reference parser.

9 Conclusion

Structural Operational Language (SOL) establishes a necessary intermediate protocol to arrest semantic drift in technical AI applications. By strictly decoupling human cognitive accelerators (SOL-H) from machine-authoritative protocol semantics (SOL-A / SOL-IR), SOL protects engineering intent across complex heterogeneous integrations. The protocol is an experimental grounding representation, designed not to bypass physical simulation, but to ensure the logic arriving at the simulator remains physically sound.

10 References

References

[1] Artsybashev, A. (2026). *SOL: Structural Operational Language — Engineering Grounding Protocol for AI-Assisted Mechanical Analysis*. Version 1.2. Kharkiv, Ukraine. Ref: AAM-V1.

[2] *ISO 281:2007*. Rolling bearings — Dynamic load ratings and rating life. International Organization for Standardization.

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